Network Security and Cryptography

11th November 2024

Examination Paper

Answer ALL questions.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination

Answer ALL questions

Marks

Question 1

a) A recent vulnerability report identifies that a critical system component is susceptible to a buffer overflow vulnerability. **4**

- Define what a buffer overflow is **and** explain its impact on the system's security and stability.

b) Explain the CIA triad in the context of information security. **6**

Total 10 Marks

Question 2

a) Describe **TWO (2)** key differences between Block and Stream Ciphers and give **ONE (1)** example of each cipher. **6**

b) Describe **TWO (2)** key differences between symmetric **and** asymmetric cryptography. **4**

Total 10 Marks

Question 3

a) Explain **THREE (3)** types of attacks that can target symmetric encryption. **6**

b) What is password salting **and** explain **THREE (3)** ways it enhances password security. **4**

Total 10 Marks

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Marks

Question 4

a) Describe **THREE (3)** common security vulnerabilities that can arise from inadequate or improper input validation in software applications. **6**

b) A user unknowingly visits a fake website and downloads and installs a file that contains malware. Describe **TWO (2)** potential impacts of the malware on the user's computer. **4**

Total 10 Marks

Question 5

a) Bob has received a confidential email from Alice, encrypted using Pretty Good Privacy (PGP). Describe the steps Bob would take to decrypt the email and verify it was indeed sent by Alice. **4**

b) Describe what an intrusion detection system (IDS) is **and** explain the difference between signature-based intrusion detection **and** anomaly-based intrusion detection. **6**

Total 10 Marks

Question 6

a) Explain what a Virtual Private Network (VPN) is **and** describe the **TWO (2)** primary types of VPNs. **6**

b) Describe **TWO (2)** common Virtual Private Network (VPN) protocols. **4**

Total 10 Marks

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Marks

Question 7

a) You are tasked with securing a small business's Wi-Fi network. Describe **THREE (3)** essential steps you would take to enhance the security of the network. **6**

b) As a security analyst at a financial institution implementing a new online banking system, describe the **TWO (2)** primary types of risk assessments you could use to evaluate potential risks. **4**

Total 10 Marks

Question 8

a) Explain what a hash function is **and** describe **TWO (2)** main uses. **6**

b) Explain **TWO (2)** examples of biometric authentication methods. **4**

Total 10 Marks

Question 9

a) Describe the difference between hardware **and** software firewalls. **4**

b) Describe **THREE (3**) weaknesses of biometric authentication. **6**

Total 10 Marks

Question 10

a) Explain why WPA2 is considered more suitable than WPA for securing wireless networks. **4**

b) Explain the concept of zero trust **and** identify **TWO (2)** key principles of zero trust. **6**

Total 10 Marks

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End of Paper

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